

Low-Loss Subwavelength Grating-Based Waveguide Crossings in Silicon Nitride at 1550 nm

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Abstract We design, fabricate, and characterise subwavelength grating waveguide crossings in SiN for 1550 nm operation. Using cascaded structures, we measure state-of-the-art 0.052 dB/crossing loss with < -47 dB crosstalk. These results enable dense, low-loss routing for telecom PICs, surpassing prior SiN crossings. ©2026 The Author(s)

Introduction

Silicon nitride (SiN) has emerged as a prominent platform for photonic integrated circuits (PICs), enabling compact, scalable, and CMOS-compatible optical systems for communications, sensing, and signal processing. Its low propagation loss, broad transparency window, and high optical power handling, make it particularly attractive for high-performance, large-scale photonic integration. Moreover, the widespread availability of multi-project wafer (MPW) platforms has facilitated broader adoption, with fabrication processes typically relying on 300 nm thick SiN. However, compared with the commonly used silicon-on-insulator (SOI) platform, its lower refractive index contrast results in weaker confinement and therefore larger passive components [1], highlighting the need for efficient and low-loss routing elements.

Among these elements, waveguide crossings are essential in dense photonic circuits, where they may contribute significantly to the overall loss, as circuit complexity and scale increases. While SOI waveguide crossings have matured significantly – evolving from geometrical designs such as elliptical intersections (<0.1 dB loss, <-25 dB crosstalk) [2], parabolic tapers (0.16 dB loss and -40 dB crosstalk) [3], mode expanders (0.028 dB loss and <-37 dB crosstalk) [4], through optimised multi-mode interferometers (MMIs) (0.007–0.4 dB loss, <-40 dB crosstalk) [5–7] to subwavelength grating (SWG) approaches (0.023–0.1 dB loss, <-40 dB crosstalk) [8, 9] – SiN crossings remain underdeveloped. The aforementioned approaches do not directly translate to SiN, and as a consequence, achieving compact and low-loss crossings in SiN, particularly at telecommunication wavelengths, remains challenging.

Available SiN waveguide crossing designs remain limited in performance, particularly at telecommunication wavelengths. An early MMI-

based implementation demonstrated broadband operation (~120 nm) and low crosstalk (<-42 dB), but suffered from relatively high insertion loss (0.65 dB for the TE polarisation) and a large footprint (~30 μm) [10]. Subsequent work reported significantly lower measured losses of ~0.05 dB per crossing and ultra-low simulated crosstalk (-60 dB) [11]. However, these results were obtained only around 840 nm and have not been experimentally validated at 1550 nm, limiting their relevance for telecom applications. In a recent inverse design approach, average losses of 0.07 dB/crossing in the telecom bands were reported, with crosstalk ~-40 dB [12], however this design is validated in 150 nm thick SiN films. Alternative approaches have explored different trade-offs between loss, footprint, and fabrication tolerance. Gradient-index (GRIN) lens-based crossings achieved moderate loss (0.46 dB), but exhibit strong sensitivity to fabrication variations (e.g. a deviation by ± 10 nm could lead to ~0.2 dB excess loss) and increased crosstalk (-29 dB) [13]. More recently, adiabatic-coupler-based designs have demonstrated improved robustness and suitability for applications such as quantum photonics, achieving losses down to 0.18 dB, albeit with relatively high crosstalk (-24.5 dB) [14]. Overall, existing approaches highlight the difficulty of simultaneously achieving low loss, low crosstalk, and compact footprint in 2D SiN waveguides at telecommunication wavelengths.

Subwavelength grating engineering offers a promising route to overcoming these limitations. By tailoring the effective refractive index through subwavelength structuring, SWGs enable enhanced modal control, broadband operation, and reduced scattering. While this approach has been successfully applied to SOI crossings, its implementation in SiN at telecommunication wavelengths remains unexplored.

In this work, we design, fabricate, and ex-

perimentally demonstrate SWG-based waveguide crossings in 300 nm thick SiN films operating at 1550 nm. Using cascaded structures we extract a crossing loss of 0.052 dB per crossing. To our knowledge, this represents the first demonstration of SWG-based waveguide crossings in SiN, as well as the lowest experimentally reported losses for planar SiN waveguide crossings at telecommunication wavelengths.

Device Design and Fabrication

The proposed waveguide crossing design consists of a SWG intersection region connected to conventional strip SiN waveguides through adiabatic SWG tapers (Fig. 1). The SWG crossing unit (highlighted in Fig. 1) was used to evaluate the insertion loss, while the layout shown in the inset was designed for crosstalk characterisation. In addition, SWG reference structures without crossings were included to isolate the excess loss introduced by the crossing, as well as straight waveguides for independent propagation loss estimation.

The crossings were implemented on a strip waveguide of width $w_{wg} = 0.95 \mu\text{m}$. To minimise mode mismatch between the strip waveguide and the SWG region, an adiabatic transition was implemented using a combined width taper and period chirp. The taper consisted of 19 grating segments, where the pitch was linearly varied from 260 nm to 350 nm, while the waveguide width was linearly reduced from $w_{t1} = 0.63 \mu\text{m}$ to $w_{t2} = 0.25 \mu\text{m}$. The length of the taper was $L_t = 11.25 \mu\text{m}$. This was followed by a uniform SWG section comprising 24 periods with a constant pitch of 350 nm and a waveguide width of $w_{t3} = 0.21 \mu\text{m}$, corresponding to a duty cycle of 50% ($a = 140 \text{ nm}$).

The crossing region itself was formed by two orthogonal SWG waveguides with identical grating parameters. At the intersection, a square segment of dimensions $1 \mu\text{m} \times 1 \mu\text{m}$ was introduced to preserve structural symmetry and ensure identical

optical conditions for both propagation directions. The dimensions of this central segment were chosen to match the effective index of the adjacent SWG sections. Following the crossing, an identical taper was used to transition back to the strip waveguide.

The device geometry was optimised using three-dimensional finite-difference time-domain (FDTD) simulations targeting minimum insertion loss at 1550 nm for the fundamental TE mode.

Crossing structures were fabricated on 300 nm thick low-pressure chemical vapour deposition (LPCVD) silicon nitride deposited on an 8-inch silicon wafer with a $2 \mu\text{m}$ buried oxide (BOX) layer. A 50 nm chromium (Cr) layer was first deposited by electron-beam evaporation to serve as a hard mask. A positive electron-beam resist (ZEP520A, $\sim 180 \text{ nm}$ thick) was then spin-coated and patterned using electron-beam lithography (EBL), followed by development in ZED-N50.

The pattern was transferred from the resist to the Cr hard mask via dry etching, and subsequently into the SiN layer using inductively coupled plasma (ICP) etching in an Oxford Instruments Plasmalab System 100 equipped with a Cobra high-density plasma source. Finally, the chips were diced to expose the waveguide facets for edge coupling and optical characterisation.

Experimental Validation

The devices were characterised using a continuous-wave (CW) tuneable laser source (TLS) over a wavelength range of 1525–1580 nm, with a constant output power of 10 dBm. Light was coupled into and out of the chip using polarisation-maintaining (PM) lensed fibres, aligned to excite the fundamental TE_{00} mode. The transmitted optical power was measured using a calibrated power sensor (PS), allowing extraction of the total insertion loss.

Propagation and coupling losses were first determined using the cutback method. Straight strip waveguides with lengths ranging from 1.5

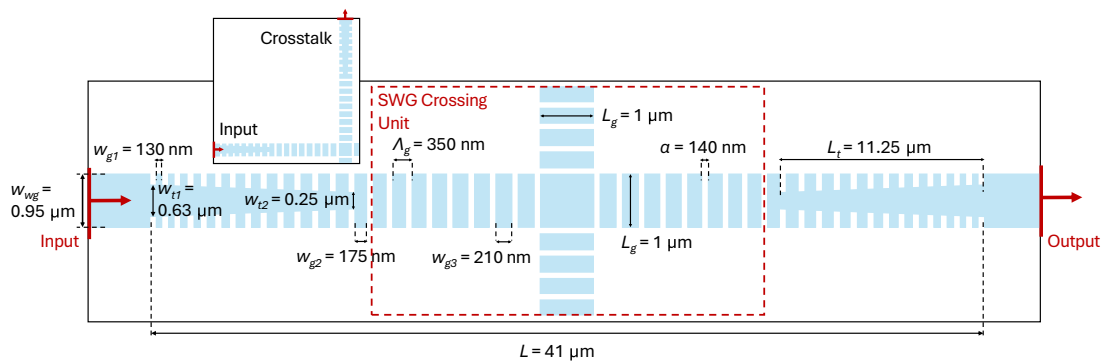


Fig. 1: Top view of a SiN waveguide crossing with input/output tapers (not to scale). The SWG crossing unit is highlighted by the red dashed square; the inset shows the crosstalk measurement layout. The solid waveguide width is w_{wg} . Taper widths are w_{t1} and w_{t2} , while grating segment widths are w_{g1} , w_{g2} , and w_{g3} at the taper start, taper end, and within the SWG crossing unit, respectively. L is the total crossing length, L_t is the taper length, L_g is grating segment length, Λ_g the grating pitch, and a the spacing between consecutive gratings.

to 5.46 cm were measured under identical coupling conditions. A linear fit of insertion loss versus waveguide length yielded a propagation loss of 1.55 dB/cm and a coupling loss of 3.12 dB per facet at 1550 nm. Across the full wavelength range, the propagation loss varied between 1.46 and 1.76 dB/cm. The SWG reference waveguides showed an excess loss of 0.066 dB relative to the solid waveguides.

The loss per crossing was evaluated using cascaded SWG structures containing $N = 1, 50, 100, 150, 200$, and 250 crossing units. For each case, identical SWG reference structures without crossings were measured to remove propagation and coupling contributions. For each value of N , three nominally identical SWG structures were measured for both the crossing and reference configurations. In some instances, measurements were repeated to mitigate alignment variability. Measurements exhibiting clear outliers, attributed to alignment errors, were excluded. The mean insertion loss of the remaining measurements was then used to calculate the excess loss due to the crossings as

$$\Delta L(N) = \bar{T}(N) - \bar{T}(0), \quad (1)$$

where $\bar{T}(N)$ and $\bar{T}(0)$ denote the mean insertion loss of structures with and without crossings, respectively.

The loss per crossing was then extracted from a linear fit of $\Delta L(N)$ as a function of N . A weighted least-squares regression was employed to account for measurement uncertainties as shown in Fig. 2. The slope of this fit corresponds to the average loss per crossing, yielding a value of 0.052 dB/crossing.

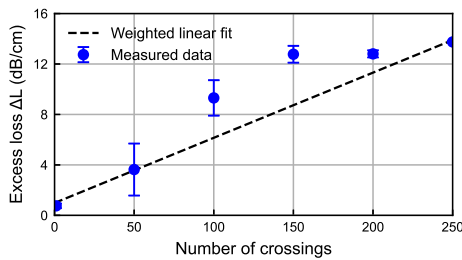


Fig. 2: Weighted linear fit of excess crossing loss to number of crossings. The slope of the fit corresponds to the average loss per crossing.

The loss per SWG taper was estimated using 3D-FDTD simulations. Devices comprising 2, 3, 4, 5, and 6 cascaded tapers were simulated, and the total insertion loss at 1550 nm was extracted in each case. A linear fit was then applied to the simulated loss as a function of number of tapers, with the slope corresponding to the average loss per taper. This analysis yielded a taper loss of 0.17 dB/taper, as shown in Fig. 3. This contribution arises from the strip-to-SWG transition and is

considered separately, as it does not form part of the intrinsic waveguide crossing loss.

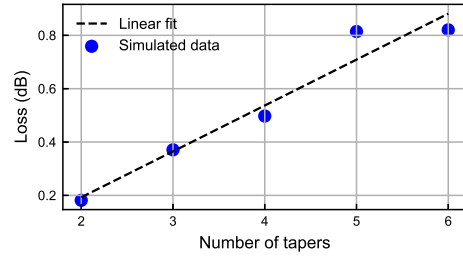


Fig. 3: Linear fit of taper loss to number of tapers. The slope of the fit corresponds to the average loss per taper.

Fig. 4a shows the wavelength dependence of the extracted loss per crossing. The loss remains approximately constant at ~0.05 dB/crossing across the measured wavelength range, with a variation of less than 0.03 dB between 1525 nm and 1580 nm. The measured loss values are of the same order as the simulated values, shown with the dashed line, indicating the broadband operation of the SWG crossings.

Crosstalk was characterised by injecting light into one waveguide of a single-crossing structure and measuring the transmitted power at the output of the orthogonal waveguide. The measured crosstalk remained below -47 dB across the wavelength range (Fig. 4b). Any divergence between the measured and simulated values can be attributed to factors such as fabrication variations and alignment errors.

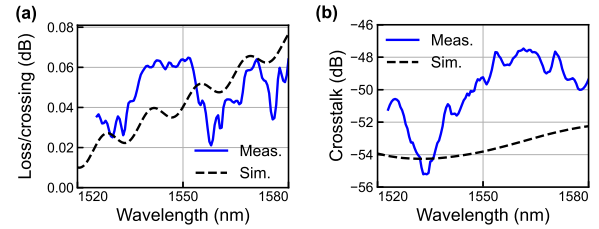


Fig. 4: Measured (solid lines) and simulated (dashed lines) (a) loss per crossing and (b) crosstalk as functions of wavelength.

Conclusions

We have designed and experimentally demonstrated subwavelength grating waveguide crossings in a 300 nm silicon nitride platform operating at telecommunication wavelengths. A loss of ~0.052 dB per crossing and crosstalk below -47 dB were measured. The loss remains nearly constant across the 1525–1580 nm range, confirming broadband performance. To our knowledge, these results represent the lowest reported loss for planar silicon nitride waveguide crossings at telecommunication wavelengths using SWG crossings in silicon nitride, highlighting their prospect as a practical solution for compact, densely interconnected silicon nitride photonic circuits.

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